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Studierichting: Informatica
Oefeningenreeks 5A nr. 8.2

$$\int_{-\frac{1}{2}}^{1} \sqrt{1-x^2}dx$$

$$\text{O.I.} = \int \sqrt{1-x^2}dx \overset{x=\sin t}{=} \int \cos^2 t dt = \int \frac{1+\cos 2t}{2}dt$$

$$= \frac{1}{2}t + \frac{1}{4}\sin 2t + c \overset{t=\text{Bgtan}\, x}{=} \frac{\text{Bgtan}\, x}{2} + \frac{\sin\left(2\,\text{Bgsin}\, x\right)}{4} + c$$

$$\Rightarrow \text{B.I.} = \left[\frac{\text{Bgtan}\, x}{2} + \frac{\sin\left(2\,\text{Bgsin}\, x\right)}{4}\right]_{-\frac{1}{2}}^{1} = \frac{\pi}{3} + \frac{\sqrt{3}}{8}$$

References